HDOC DAY – 52

Algorithm: Previous Greater Element

1. Start.
2. Read the size of the array size.
3. Check whether size is positive.
4. Declare a variable-length integer array arr[size].
5. Read all elements of the array.
6. Pass the address of arr[] to the function previous_greater().
7. In previous_greater():
    - Repeat for every element arr[i].
    - Set previous = -1.
    - Start an inner loop from j = i - 1 and move towards index 0.
    - Compare arr[j] with arr[i].
    - If arr[j] > arr[i]:
        - Store arr[j] in previous.
        - Use break because the nearest greater element on the left has been found.
    - If no greater element is found, previous remains -1.
    - Display previous.
8. Separate each result using commas.
9. Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
void previous_greater(int *arr, int size)
{
   int i, j, previous;

   for (i = 0; i < size; i++)
   {
      previous = -1;

      // Search elements to the left
      for (j = i - 1; j >= 0; j--)
      {
         if (*(arr + j) > *(arr + i))
         {
            previous = *(arr + j);
            break;  // Nearest greater element found
         }
      }

      printf("%d", previous);

      if (i < size - 1)
      {
         printf(", ");
      }
   }
```

```
    printf("\n");
}

int main()
{
    int size, i;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid size.\n");
        return 0;
    }

    int arr[size];

    printf("Enter the array elements:\n");

    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }
```

```
    // Pass the address of the array
    previous_greater(arr, size);

    return 0;
}
```

Test Case Table

Test Case	Size	Input Array	Expected Output
1	4	4 5 2 25	-1, -1, 5, -1
2	4	13 7 6 12	-1, 13, 7, 13
3	4	1 3 2 4	-1, -1, 3, -1
4	5	1 2 3 4 5	-1, -1, -1, -1, -1
5	5	5 4 3 2 1	-1, 5, 4, 3, 2